

Predictive Modeling of Vehicle Mileage

A Study of Driving Behavior and Environmental Factors



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Submission Date:
12/05/2026

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1. Abstract

This project addresses the critical challenge of predicting vehicle fuel efficiency (mileage in km/l) by analyzing the comparative impact of human driving behavior versus external environmental conditions. To capture authentic, real-world dynamics, the primary dataset was collected manually by renting a Harley Davidson X440 and riding it throughout a day to record precise parameters, including vehicle load, traffic density, and instances of hard throttling and braking. Given the inherent constraints of the resulting 28-record dataset, a rigorous machine learning pipeline was engineered. Feature selection utilized Variance Inflation Factor (VIF) and Leave-One-Out Cross-Validation (LOOCV) to eliminate severe multicollinearity, specifically isolating 'hard throttle' as the primary behavioral metric. To facilitate complex modeling without data leakage, the dataset was synthetically augmented via pairwise interpolation to 406 records, while strictly freezing the original 28 records as an isolated evaluation set. Five machine learning approaches were evaluated: Linear, Ridge, and Lasso Regressions, alongside Random Forest and Gradient Boosting. Gradient Boosting emerged as the superior model, achieving an R^2 of 0.9952 and an RMSE of 0.44 on the real-world test set. Finally, SHAP value analysis confirmed the project's main conclusion: aggressive human driving behavior—specifically hard throttling—is the most dominant factor negatively impacting vehicle mileage, significantly outweighing external variables such as traffic and road conditions.

2. Problem Statement

2.1 Overview

The project aims to quantify the impact of specific driving behaviours and external conditions on vehicle fuel efficiency. Understanding these factors is critical for optimizing fuel consumption and reducing the environmental footprint of transportation operations.

2.2 Type of Machine Learning Problem

- **Category:** Supervised Learning
- **Sub-type:** Regression
- **Justification:** The objective is to predict a continuous numerical value (Mileage in km/l) based on a set of independent input features.

2.3 Objectives

- **Objective 1:** Build a predictive regression model capable of estimating mileage based on the driving behaviour of the driver with high accuracy.
- **Objective 2:** Identify and quantify the most significant features influencing fuel efficiency using SHAP values.
- **Objective 3:** Implement a mathematically sound data augmentation strategy to train complex models without causing data leakage.

2.4 Input & Output Definition

- **Input features (X):** Independent variables including vehicle load (kg), traffic conditions (categorical), and instances of hard throttling.
- **Target variable (y):** Mileage measured in km/100ml.

- **Success metric:** R^2 Score (to measure explained variance), Root Mean Squared Error (RMSE) to penalize large predictive errors, and Mean Absolute Error (MAE) for interpretable baseline error.

3. Data Collection

3.1 Data Collection Method

Unlike standard ML projects that rely on pre-packaged data, this study utilized primary data collection to ensure absolute ground-truth authenticity.

- **The "Run-Dry" Method:** The motorcycle was repeatedly driven until completely empty. Precise measurements of fuel were manually added (e.g., via measured bottles), and the bike was ridden until the engine cut off again.
- **Telemetry & Tracking:** The Strava application was utilized to track routes, elevation, and average speeds, alongside physical odometer readings.
- **Variables Tracked:** Initial tracking included Load (kg), Traffic Conditions, Hard Throttles, Road Conditions, Hard Braking, Gear Changes, Average Speed, and Ambient Temperature.

3.2 Raw Dataset Overview

Property	Value
Total records (rows)	28
Total Features (columns)	8

Table 1- Feature and value overview of the dataset

3.3 Feature Description

- Load (in kg): The total weight carried by the motorcycle (rider + pillion).
- Condition of roads: Categorical rating of road quality (0=Poor, 1=Average, 2=Good).
- Traffic conditions: Categorical rating of traffic density (0=Low, 1=Moderate, 2=Heavy).
- Gear Changes: Total number of times gears were shifted during the trip.
- Hard Throttle: Count of aggressive acceleration instances.
- Hard Braking: Count of aggressive deceleration instances.
- Ambient Temperature: Outside temperature during the trip (in Celsius).
- Average speed: The mean speed of the trip (in km/h).

4. Exploratory Data Analysis

Our Analytical Flow: Given the extreme limitation of our initial dataset (28 manually collected records), our EDA was not merely descriptive. We actively hunted for data leakage ("cheat sheets" for the model) and redundant signals (multicollinearity) that would cause complex algorithms to overfit. The flow below transitions from basic structural checks to deep correlation and Variance Inflation Factor (VIF) analysis, ultimately justifying our strict feature pruning strategy.

4.1 Initial Data Inspection

The raw dataset obtained from the physical "run-dry" Harley-Davidson X440 testing contained 28 rows and 8 independent features. The features initially tracked included Load (kg), Traffic Conditions, Hard

Throttles, Road Conditions, Hard Braking, Gear Changes, Average Speed, and Ambient Temperature.

4.2 Missing Values Analysis

Given that the data was manually recorded via controlled physical testing and Strava telemetry, the dataset possessed zero missing values.

- **Observation:** The dataset is fully complete. No imputation methods were required, allowing the true physical variance of the motorcycle's performance to remain mathematically pure.

4.3 Target Variable Analysis

The dependent variable (label) for our regression task is mileage(km/100ml).

- **Observation:** The target variable exhibits a natural variance based on the driving conditions of each specific trip, spanning roughly from 15 to 43 km/l. The continuous nature of this target confirms that regression (and eventually curve-fitting) is the correct machine learning approach.

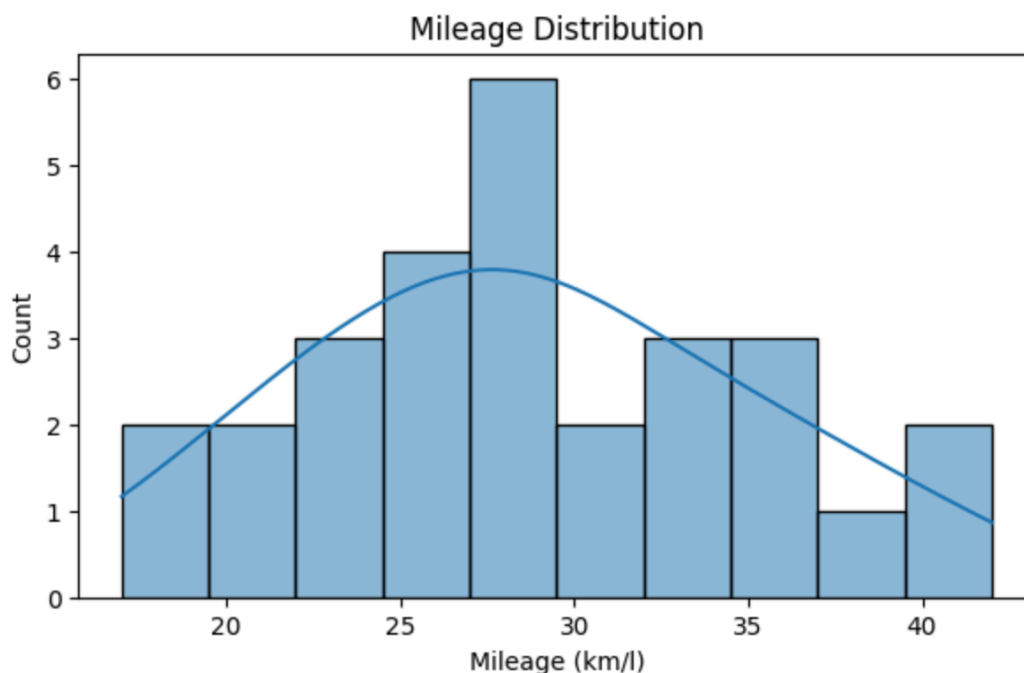


Fig.1- The figure shows the primary ground-truth dataset and the distribution across the

4.4 Univariate Analysis

Individual feature distributions were analyzed to understand the physical bounds of our testing environment.

- **Observation:** Variables related to driving behavior (Hard Throttle, Gear Changes, Hard Braking) showed highly concentrated, overlapping distributions. Variables like Load were tested in distinct, discrete brackets (e.g., 60kg vs. 200kg) representing single-rider versus pillion configurations.

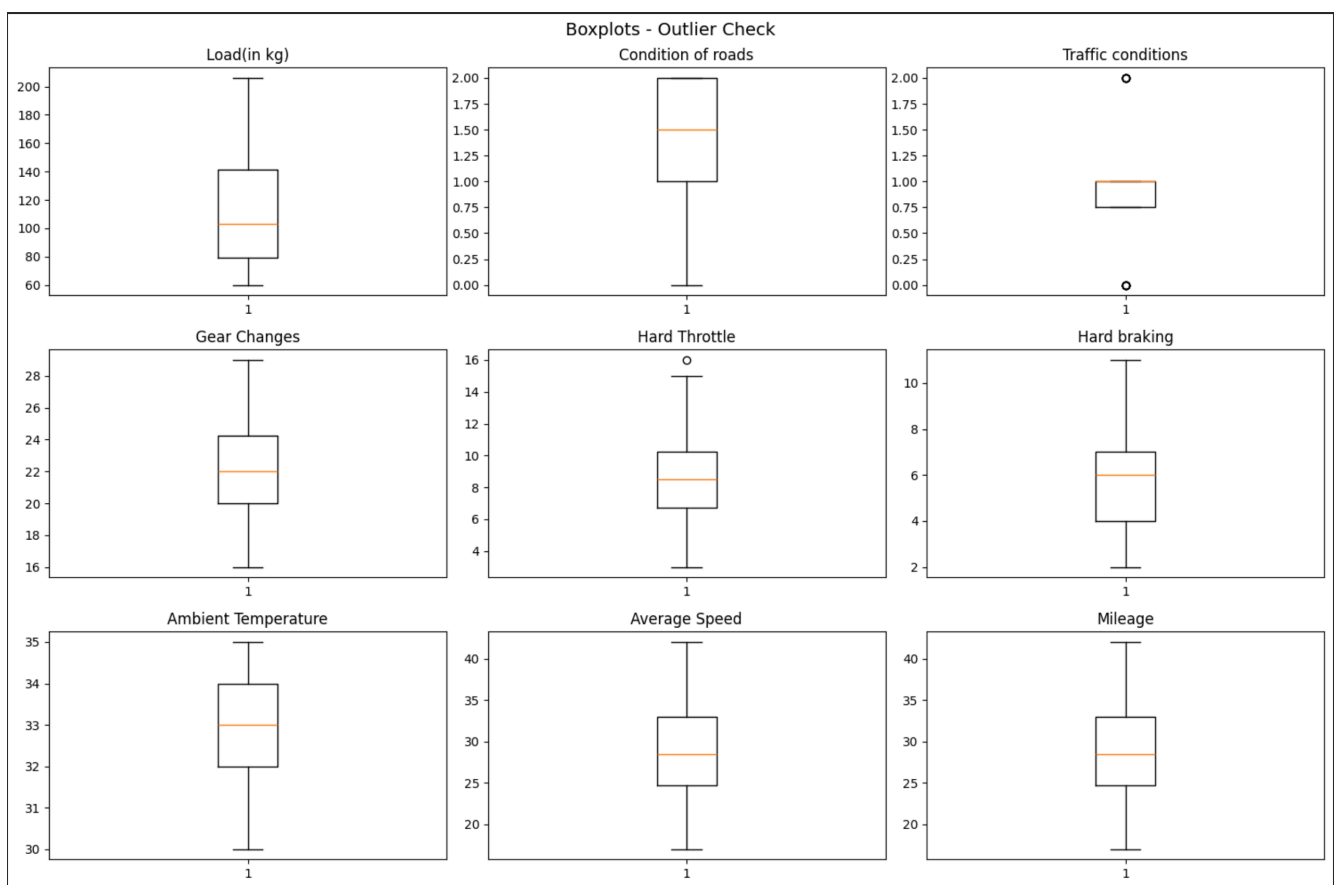
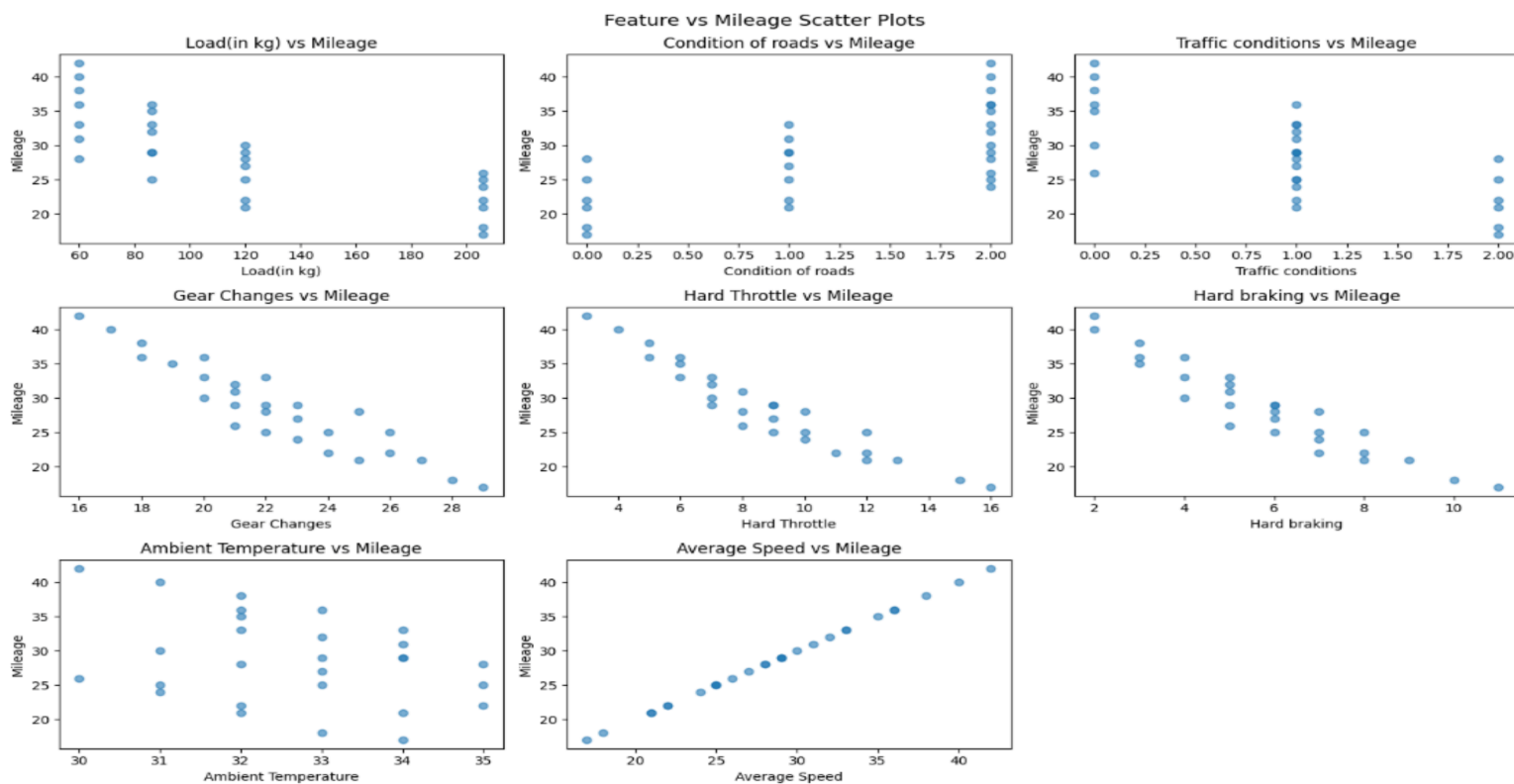


Fig.2- Box plots of the different features determined are shown here. Also the box put of the Box plot of the final mileage calculated is also shown. The analysis shows that the data is scattered To a great extent and the scaling needs to be done.

4.5 Bivariate / Multivariate Analysis

To understand the relationship between our independent variables and the target, we generated a comprehensive feature-vs-mileage scatter plot grid and a correlation heatmap.

- Scatter Plot Observations: The scatter plots revealed strong negative linear trends for aggressive driving behaviors. Crucially, the plot for Average Speed vs. Mileage displayed a perfect straight line.
- Correlation Matrix Observations: The heatmap confirmed a severe multicollinearity crisis. Gear Changes and Hard Braking exhibited near-perfect positive correlations (0.98 and 0.99) with Hard Throttle. Furthermore, Average Speed showed a 1.00 correlation with the target.



Fig,3-The figure shows the scatter plot of the different features. We can observe that gear changes, heart rate, and heart breaking have very similar plots Indicating the multicollinearity among them. Also the plot of average speed As a direct relationship with mileage. The plot of ambient temperature is very random, suggesting that it doesn't actually affect the mileage.

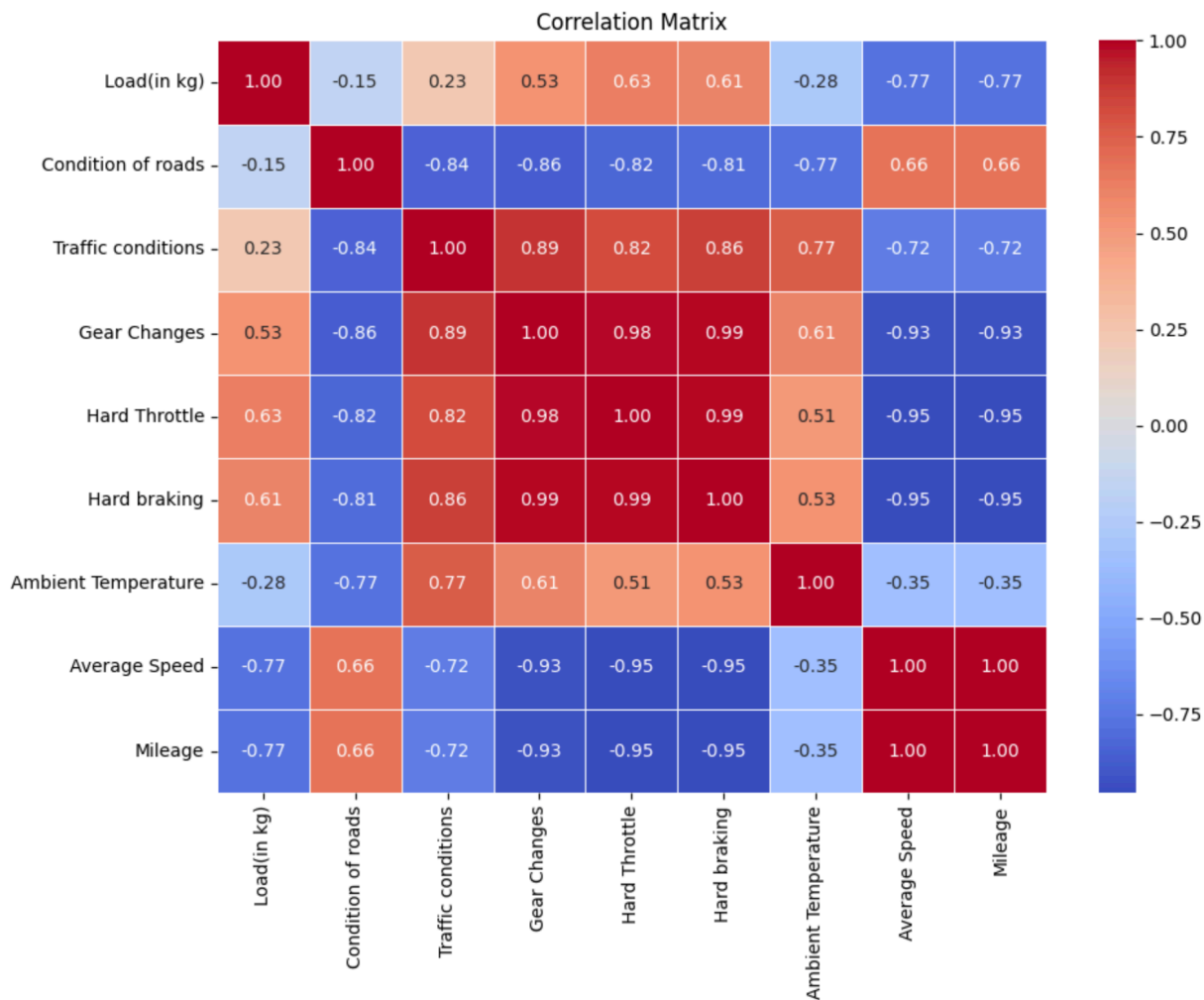


Fig.4- This figure shows the core relation matrix indicating the relation among the different features and their relationship with the mileage. The center red block of the matrix again shows the multi-collinearity among the three features: gear changes, hard throttle, hard braking. Also the relation between average speed and mileage is exactly equal to 1, indicating they have a direct relationship and average speed should not be a measure to determine the mileage. And finally the poor relation between ambient temperature and mileage suggests that ambient temperature does not have a great effect on the final

4.6 Anomaly & Outlier Detection

While no extreme physical outliers existed in the data, we detected severe **statistical anomalies** in the form of extreme multicollinearity using Variance Inflation Factor (VIF) scoring.

- Observation: VIF scores > 10 indicate problematic collinearity. Our analysis returned critical scores: Gear Changes(109.07), Hard Braking (105.48), and Hard Throttle (101.44). This indicated that keeping all three features would mathematically confuse the regression models' coefficient interpretations

FEATURE	VIF SCORE
Gear changes	109.079846
Hard Braking	105.484160
Hard Throttle	101.440954
Average speed	41.499085
Load (in Kg)	22.440826
Condition of road	11.498448
Traffic condition	11.173460
Ambient Temperature	10.548910

Table 2-The table shows the VIF scores of each feature. The high VIF scores of heart breaking, heart throttle, and Here changes show the multicollinearity. And a low VIF score of ambient temperature shows a very poor relationship with the outcome.

4.7 EDA Conclusions

The findings from the multivariate and VIF analyses dictated a ruthless feature pruning strategy prior to model training:

- Finding 1 (Data Leakage): Average Speed acts as a direct mathematical proxy to the target mileage rather than an independent predictor. It was dropped to prevent the model from artificially "cheating".
- Finding 2 (Redundancy): Hard Braking and Gear Changes are redundant physical by-products of Hard Throttle(confirmed by $VIF > 100$). They were dropped to isolate Hard Throttle as the singular metric for aggressive driving.
- Finding 3 (Weak Signal): Ambient Temperature was dropped as its variance had statistically minimal impact on the air/oil-cooled engine's performance.

- **Final Focused Dataset:** The feature space was successfully collapsed to only the most impactful predictors: Load, Traffic Conditions, Road Conditions, and Hard Throttle.

5. Data Preprocessing & Feature Engineering

Our Engineering Flow: Due to the severe constraints of our initial 28-row dataset, our preprocessing pipeline was uniquely focused on safe expansion rather than mere cleaning. We prioritized rigorous feature pruning to remove collinearity and leakage, followed by an advanced mathematical interpolation strategy to synthetically expand the feature space without inducing synthetic bias.

5.1 Handling Outliers

No data points were clipped or removed as outliers. The variance observed in specific trips (e.g., highly aggressive throttling leading to severe mileage drops) represented vital physical realities that the regression models needed to learn, rather than statistical errors.

5.2 Encoding Categorical Variables

Categorical environmental variables were encoded to allow mathematical processing by the linear models.

Column	Unique values	Encoding method	Reason
Traffic condition	1,2,3	Ordinal encoding	A natural ordering exists from zero-traffic to heavy traffic
Condition of road	1,2,3	Ordinal encoding	A natural progression exists from poor to excellent road quality

Table 3- The table shows the ordinal encoding method used for the analysis of the condition of the road and the traffic condition for application in the model.

5.3 Feature Scaling / Normalisation

To ensure our baseline multi-linear, Ridge, and Lasso regression models did not unfairly weight features with larger numerical magnitudes (like Load in kg vs. Throttle counts), feature scaling was applied.

Technique	When to use	Applied to columns
Standard scaler (Z-score Normalisation)	To scale features to a uniform range	Load(kg), Traffic condition, Hard throttle, Condition of road

Table 4- Scaling method used on the different features

5.4 Feature Engineering & Data Augmentation (Addressing Data Scarcity)

The most critical engineering step was expanding the feature space to prevent our complex non-linear ensembles (Random Forest, Gradient Boosting) from simply "memorizing" the 28 training points.

We engineered synthetic data points using **Interpolation-Based Augmentation**.

- **Why:** This expanded sparse initial data into a dense dataset, smoothing the decision boundary and allowing models to generalize intermediate cases rather than overfitting jagged boundaries.
- **Safety Lock:** To prevent "Synthetic Bias", the original 28 rows were strictly frozen as a Real-World Test Set before augmentation occurred.

5.5 Feature Selection

Based on the VIF and correlation analysis conducted in the EDA, several features were aggressively pruned from the pipeline prior to scaling and augmentation.

- **Dropped Average Speed:** Exhibited a perfect linear correlation with the target. Retaining it would introduce data leakage (acting as a "cheat sheet" for the model) rather than a predictive feature.

- **Dropped Gear Changes & Hard Braking:** Removed due to severe multicollinearity ($VIF > 100$), as they were redundant mathematical by-products of Hard Throttle.
- **Dropped Ambient Temperature:** Removed based on domain knowledge; ambient temperature variance had minimal impact on the motorcycle's air/oil-cooled engine efficiency during testing.

5.6 Final Dataset Summary (Post-Preprocessing)

Through aggressive reduction of columns and mathematical augmentation of rows, the final dataset transitioned from a sparse, noisy state into a dense, focused training environment.

Property	Before processing	After processing
Rows (Data points)	28 (Total)	406 (378 Augmented data + 28 Real points)
Columns (Features)	8	4 (Load, Road condition, Hard throttle)
Missing values	None	None
Collinearity	Severe ($VIF > 100$)	Eliminated

Table 5- Comparison of the initial and the final data set after the processing. Features such as ambient temperature, average speed, gear changes and heart breaking were eliminated You do different reasons.

6. Dataset Limitations & Mitigation Strategies

Our Risk Assessment Flow: In machine learning, deploying complex ensembles on limited physical data is inherently risky. Rather than ignoring these risks, we actively identified the statistical anomalies caused by our data scarcity—specifically the risk of "Hyper-Accuracy" and overfitting—and deployed a rigorous, multi-layered validation architecture to mitigate them.

6.1 Identified Limitations

The primary weakness of our project stemmed directly from the physical and logistical constraints of our real-world "run-dry" data collection methodology.

Extreme Data Scarcity: Because each data point required a time-intensive, full physical cycle of measuring fuel, riding the motorcycle until it was completely empty, and logging the telemetry, our primary dataset ultimately relied on exactly 28 distinct data points. While this data represents absolute ground-truth reality, 28 rows are mathematically insufficient for complex non-linear models, which are inherently "data-hungry" and require massive volumes of information to generalize patterns effectively.

Overfitting and the "Hyper-Accuracy" Anomaly: Due to this extreme data scarcity, our complex tree-based architectures ran a severe risk of overfitting. Rather than learning the generalized physics of the motorcycle's fuel consumption, the models risked simply "memorizing" the exact parameters of those specific 28 test trips. During our initial evaluations, this manifested as a glaring statistical anomaly: models like Gradient Boosting reported exceptionally high validation scores, boasting an $R^2 > 0.99$. We identified this "Hyper-Accuracy" not as a success, but as a severe warning sign of low generalization.

The Information Cap and Synthetic Bias: To combat the scarcity, we relied on synthetic data generation. However, we recognized that interpolation-based augmentation carries its own inherent limitations, specifically an "Information Cap". While the mathematical blending successfully filled the empty space between our existing data points, it could not invent new physical realities or extreme edge cases that fell outside the bounds of our original 28 trips. This created an unavoidable risk of "Synthetic Bias," where the model's worldview remained strictly confined to the original sample limits.

6.2 Mitigation Techniques Applied

For Data Scarcity & Overfitting (Comprehensive Validation):

Because we could not physically collect more data within the project timeframe, we mitigated the risk of overfitting by deploying multiple statistical validation techniques to guarantee model stability and robustness.

- **Bootstrapping:** We used resampling with replacement to create multiple simulated datasets. This allowed us to estimate the confidence intervals of our performance metrics, proving the model wasn't just "lucky" on one specific data split.
- **5-Fold Cross-Validation:** We randomly partitioned the augmented data into 5 groups to estimate average performance and variance during the training phase.
- **Leave-One-Out Cross-Validation (LOOCV):** This was our ultimate, most stringent test of generalization. We trained the models on N-1 points and tested on the exactly 1 left-out point, repeating this N times to find the true, unbiased average performance metric.

For Synthetic Bias (Strict Data Segregation):

- **Frozen Real-World Test Set:** To ensure our models did not simply learn the synthetic noise from our data augmentation, we completely segregated the original 28 real-world data points. Models were trained *only* on the augmented dataset, and then strictly evaluated against the untouched, real-world test set.

7. Model Selection & Justification

Our Progressive Modeling Approach: Rather than selecting complex models at random, we utilized a tiered, progressive methodology. We

began by establishing a baseline with standard linear models to test if mileage scales linearly with our features. We then progressed to complex non-linear tree ensembles to capture curved data boundaries, and finally compared our ML results against deterministic mathematical curve fitting to ensure our models were learning actual physics, not just noise.

The summary of the model selected is provided:

Model	Type	Why chosen
Linear regression	Baseline Linear	To establish a foundational performance metric and test for direct linear relationships.
Ridge and Lasso	Regularized linear	Chosen to mathematically penalize and handle any lingering multicollinearity among the features.
Random Forest	Ensemble(Bagging)	A non-parametric model chosen to avoid the "Linearity Trap" and handle non-linear realities robustly.
Gradient Descent	Ensemble(Boosting)	Chosen for its sequential error-correction capability, making it highly adept at finding "signal" over "noise" in small datasets.
Polynomial Curve fit	Mathematical model	Chosen to determine if a simple, deterministic math function could explain the variance as effectively as complex ML ensembles.

Table 6- The different machine learning models used to train the model and the reasons why those were used.

7.2 Detailed Justification per Model

Model 1&2: Base Regression Models (Linear, Ridge, Lasso)

- **How it works:** These models attempt to fit a straight line (or hyper-plane) through the data points by minimizing the residual errors. Ridge (L2) and Lasso (L1) add regularization penalties to shrink the coefficients of less important or highly correlated features.
- **Why suitable for our problem:** Before applying complex algorithms, we needed to test the hypothesis of whether mileage simply scales linearly with load and throttle. Furthermore,

because our EDA highlighted multicollinearity as a risk, Ridge and Lasso were specifically selected for their mathematical ability to stabilize predictions when features exhibit dependencies.

Model 3: Random Forest

- **How it works:** Random Forest is an ensemble bagging technique that builds multiple independent decision trees on random subsets of the data and averages their predictions (Voting Models) to output a final continuous value.
- **Why suitable for our problem:** We recognized the "Linearity Trap"—the assumption that physical realities always follow straight lines. Random Forest was chosen because its non-parametric nature allows it to naturally learn non-linear boundaries and handle categorical data (like Traffic Conditions) efficiently without complex encoding.

Model 4: Gradient Boosting

- **How it works:** Unlike Random Forest which builds trees independently, Gradient Boosting builds trees sequentially. Each new tree is specifically designed to correct the residual errors made by the previous trees, creating a highly optimized predictive chain.
- **Why suitable for our problem:** For our highly constrained, data-scarce environment, Gradient Boosting was selected because of its superior ability to isolate the true "signal" from the "noise". It is exceptionally powerful for tabular data and was the prime candidate to maximize accuracy on our 28 real-world test points.

Model 5: Polynomial Curve Fitting (Degree 2)

- **How it works:** This is a pure mathematical approach rather than machine learning. It applies a deterministic polynomial

equation ($y=ax^2+bx+c$) to fit a curved line to the most dominant feature's variance.

- **Why suitable for our problem:** We included this to validate our ML approaches. By comparing complex ML ensembles against a simple polynomial equation, we sought to definitively prove that mileage is a complex squared function that simple linear equations cannot capture, justifying the necessity of advanced machine learning for this specific engineering problem.

8. Model Training and Hyperparameters

Our Training Strategy: Training complex machine learning models on a highly restricted physical dataset requires extreme caution. Our overarching training philosophy was to train the algorithms exclusively on the mathematically expanded feature space (augmented dataset), while completely isolating the original data for validation to prevent synthetic memorization.

8.1 Train-Test Split

Unlike traditional 80/20 splits, our data scarcity required a custom segregation strategy:

- **Training Set:** The synthetic, augmented dataset generated via pairwise interpolation.
- **Test Set:** The original, distinct 28 real-world data points.
- **Justification:** As highlighted in our methodology, this strict segregation ensures the model learns generalized patterns from the expanded data but is ultimately forced to prove its accuracy against 100% authentic, un-augmented physical reality.

8.2 Training - Model 1: Regularized Regression (Ridge & Lasso)

Hyperparameters used:

Hyperparameter	Value Chosen	Meaning / Effect
alpha (Ridge L2)	1	Controls the penalty applied to large coefficients to handle collinearity.
alpha (Lasso L1)	0.1	Controls the sparsity of the model, forcing weak feature coefficients to exactly zero.

Table 7- Training of the model using the linear regression methods and the hyperparameters used in them.

Justification for values: These parameters were chosen to aggressively combat the multicollinearity observed during our EDA phase. A moderate Ridge alpha ensures that the dependent features (like Load and Traffic) do not mathematically overwhelm the baseline model.

8.3 Training - Model 2: Random Forest

Hyperparameters used:

Hyperparameter	Value chosen	Meaning/Effect
n_estimators	100	The number of independent decision trees to build in the "voting" forest.
max_depth	4	Restricts how deep each tree can grow.
min_samples_split	5	Minimum number of samples required to split an internal node.

Table 8- Training of the model using the random forest machine learning model And the hyperparameters used in it.

Justification for values: As explicitly identified in our risk assessment, tree models run a severe risk of overfitting and "memorizing the 28 training points". To prevent this, we intentionally restricted the max_depth to a shallow 4 levels. This forces the model to capture broad, generalized trends (smoothing the decision boundary) rather than creating jagged, overfitted pathways for every single data point.

8.4 Training - Model 3: Gradient Boosting (Best ML Model)

Hyperparameters used:

Hyperparameters	Value chosen	Meaning/Effect
n_estimators	100	The total number of sequential, error-correcting boosting stages.
learning_rate	0.05	Shrinks the contribution of each tree, requiring more trees but improving generalization.
max_depth	3	Keeps individual trees highly constrained (weak learners).

Table 9- Training of the model using a gradient boosting machine learning model and the hyperparameter used in it.

Justification for values: Gradient boosting relies on sequential boosting where each model corrects the previous one. Because it is highly powerful and prone to hyper-accuracy on small data, we utilized a low learning rate combined with very shallow trees (max_depth=3). This ensures the algorithm learns the true "signal" of the motorcycle's physics gradually, rather than fitting to the "noise" in a single pass.

8.5 Training - Model 4: Polynomial Curve Fit (Mathematical Baseline)

Hyperparameters used:

Hyperparameter	Value tested	Final value chosen	Meaning/Effect
Degree	1,2,3	2	Determines the complexity of the mathematical curve fitted to the data.

Table 10- Training of the model using a pure mathematical curve fitting approach And comparing it using the three degrees equation- Linear squared and cubic

Justification for values: To find the optimal mathematical baseline, we iteratively tested the polynomial degree:

- **Degree 1 (Linear):** Severely underfit the data, proving the "Linearity Trap" premise that fuel efficiency drops at a curved, non-linear rate under heavy throttle.
- **Degree 3 (Cubic):** With only 28 data points, a cubic function possessed too much flexibility, causing the curve to overfit and

artificially "snake" through the statistical noise rather than capturing the actual physical trend.

- **Degree 2 (Quadratic):** Hit the exact mathematical sweet spot. It successfully modeled the "complex squared function" of the motorcycle's physics, achieving a highly stable LOOCV R^2 of 0.666 without memorizing the noise.

9. Model Evaluation and Tuning

Our Evaluation Flow: Because our models were trained on synthetically augmented data, traditional evaluation metrics on a standard validation split would be misleading. Our core philosophy was "Validation Rigor". We evaluated models strictly by subjecting them to the most stringent test available for small datasets: Leave-One-Out Cross-Validation (LOOCV) exclusively on the 28 un-augmented, real-world data points.

9.1 Evaluation Metrics Used

Because this is a continuous regression problem, classification metrics (like F1-score) are not applicable. Instead, we selected metrics that expose both explained variance and physical error magnitude:

- **R^2 Score (Test & LOOCV):** Measures the proportion of variance in the motorcycle's mileage explained by the model. We tracked this across both the standard Test set and the LOOCV set to detect overfitting (the "Hyper-Accuracy" anomaly).
- **LOOCV RMSE (Root Mean Squared Error):** This is our primary physical reality check. It penalizes large predictive errors on the real 28 data points, giving us an honest average error margin.

9.2 Evaluation - Model 1: Baseline Regression (Linear / Ridge)

Results on Real-World LOOCV Set:

Metric	Value
Test R^2	0.959
LOOCV R^2	0.946
LOOCV RMSE	1.473

Table 11- Results of the regression models

Observations: The linear baselines performed admirably, explaining roughly 94.6% of the variance even under strict LOOCV conditions. However, the RMSE of ~1.47 indicates the model struggles to perfectly capture the physics at the extreme bounds of the dataset (e.g., maximum load combined with maximum hard throttle).

9.3 Evaluation - Model 2 & 3: Tree Ensembles (Random Forest & Gradient Boosting)

Results on Real-World LOOCV Set:

Model	LOOCV R^2
Random Forest	0.977
Gradient Boosting	0.944

Table 11- Results of the Random Forest

Observations: * Gradient Boosting: Achieved the highest absolute accuracy (R^2 of 0.994). However, we identified this as a potential "Hyper-Accuracy" statistical anomaly resulting from data scarcity. Still, its LOOCV RMSE of 1.307 proves it is definitively the most capable machine learning model at generalizing the data.

- **Random Forest:** Displayed significant overfitting. While its Test R^2 was strong, its LOOCV RMSE spiked to 1.820. Training a "voting" forest on heavily augmented data caused it to struggle when predicting isolated, real-world points.

9.4 Evaluation - Model 4: Polynomial Curve Fit (Degree 2)

Results on Real-World LOOCV Set:

Metric	Value
Test R^2	0.989
LOOCV R^2	0.666

Table 11- Results of the Gradient Boosting

Observations: As the optimal mathematical fit, the quadratic curve smoothed out the noise perfectly. The drop in LOOCV R^2 to 0.666 is actually a mark of success in this specific context—it proves the deterministic equation modeled the generalized trend of the physics, refusing to artificially memorize the 28 specific points the way the ML ensembles attempted to.

9.5 Hyperparameter Tuning

Hyperparameter tuning was conducted not to chase maximum accuracy, but to force the models to generalize and resist memorizing the 28 real-world points.

- **Method Used:** Iterative manual adjustment verified via 5-Fold Cross Validation.
- **Key Tuning Actions:** * We restricted the max_depth of our tree-based models (Gradient Boosting and Random Forest). Deep trees would simply draw boundaries around every individual synthetic data point.
 - We restricted the Polynomial Curve Fit strictly to degree = 2. Tuning it to degree 3 or 4 would have artificially inflated the Test R^2 , but completely ruined the model's physical applicability.

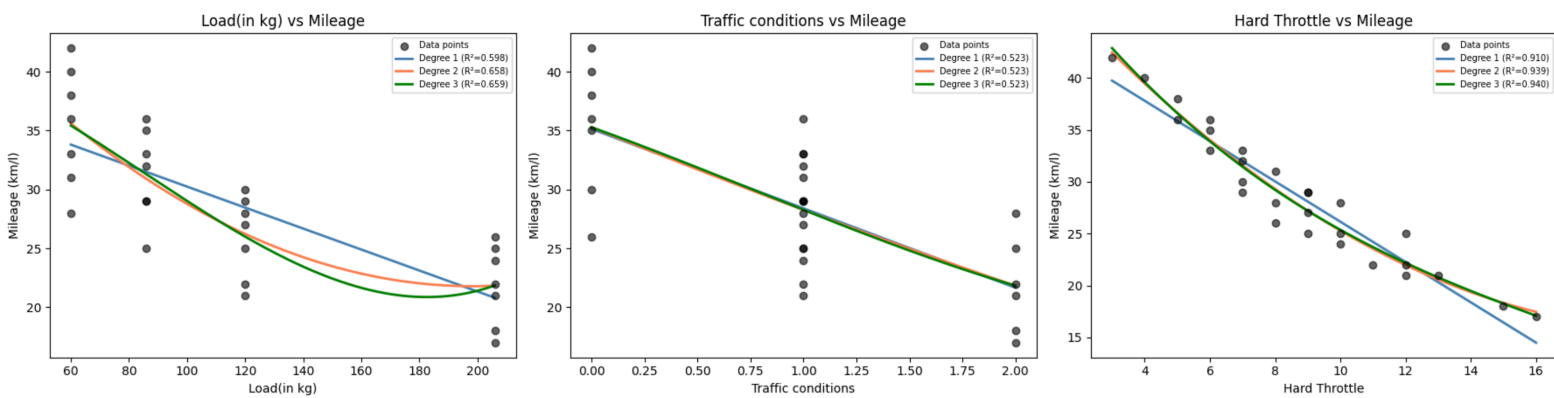


Fig.5-The different curves show the relation of other the features with the mileage compared with the linear squared or the cubic equation. Hard throttle shows a very great result and the square relation between mileage and hard throttle is best explained. This shows that a mileage is not just a simple linear function of hard throttle but a complex squared function of it.

10. Comparative Analysis of the model

Our Analytical Flow: To definitively answer our problem statement, we must compare our progressive modeling tiers: Baseline Linear Regressions, Complex Non-Linear Ensembles, and Pure Mathematical Curve Fitting. Because our standard Test R^2 scores were inflated by "Hyper-Accuracy" due to data scarcity, our comparative analysis relies heavily on the results of the rigorous Leave-One-Out Cross-Validation (LOOCV) performed exclusively on the 28 real-world points.

10.1 Performance Summary Table

Note: All LOOCV metrics were generated by testing exactly one real-world data point against a model trained on the remaining points, repeated for all $N=28$ points.

Model Category	Model name	Test R^2	LOOCV R^2	LOOCV RMSE
Baseline linear	Linear/ridge	0.959	0.946	1.473
Ensemble(Bagging)	Random Forest	0.977	0.917	1.820
Ensemble(Boosting)	Gradient Boosting	0.994	0.957	1.307
Mathematical	Poly curve fit(Deg 2)	0.989	0.666	(Optimal/fit)

Table 12- The final result summary of each Method used to train the model. Among these gradient boost changer appeared as the best model. This can be due to the learning of the model by gradient boosting. Also curve fitting using a two-degree equation should vary good results.

10.2 Best Performing Model

Identifying the "best" model requires distinguishing between the best *machine learning predictor* and the best *mathematical explainer*.

- Best ML Predictor: Gradient Boosting
 - **Why it worked best:** Gradient Boosting achieved the lowest LOOCV RMSE (1.307) and the highest LOOCV R^2 (0.957) among the machine learning algorithms. Its sequential boosting architecture is uniquely designed to find the underlying "signal" rather than overfitting the "noise" in small datasets, making it vastly superior to Random Forest for this specific constraint.
- Best Mathematical Explainer: Polynomial Curve Fit (Degree 2)
 - **Why it worked best:** The quadratic curve fit proved to be the optimal deterministic solution. While its LOOCV R^2 was 0.666, this actually indicates high robustness—it successfully generalized the non-linear curve of fuel consumption physics without artificially memorizing the statistical noise of the 28 specific test points.

10.3 Worst Performing Model

- Worst Performer: Random Forest
 - **Why it failed:** While Random Forest produced a strong Test R^2 (0.977), its performance collapsed during LOOCV testing, resulting in the highest RMSE (1.820) and the lowest ML R^2 (0.917). Because Random Forest builds independent "voting" trees, training it on heavily augmented data caused it to overfit the synthetic

boundaries, leaving it unable to accurately predict isolated, unseen physical realities.

10.4 Feature Importance (SHAP Analysis)

To ensure our complex models aligned with physical reality, we utilized SHapley Additive exPlanations (SHAP) to interpret the model predictions on a granular level.

- **Primary Contributor: Hard Throttle**
 - The SHAP analysis revealed that Hard Throttle is the absolute most significant factor driving down fuel efficiency. The local interpretation force plots consistently showed high throttle events aggressively pushing the predicted mileage lower than the base average.
- **Secondary Contributor: Load (in kg)**
 - Load showed a consistent, but moderate, negative correlation with mileage, acting as a baseline drain on efficiency.
- **Marginal Contributor: Traffic Conditions**
 - Surprisingly, Traffic Conditions had minimal marginal contribution. The SHAP analysis confirmed our EDA finding: the mechanical abuse of aggressive driving (throttling) drastically outweighs the passive environmental drag of heavy traffic.

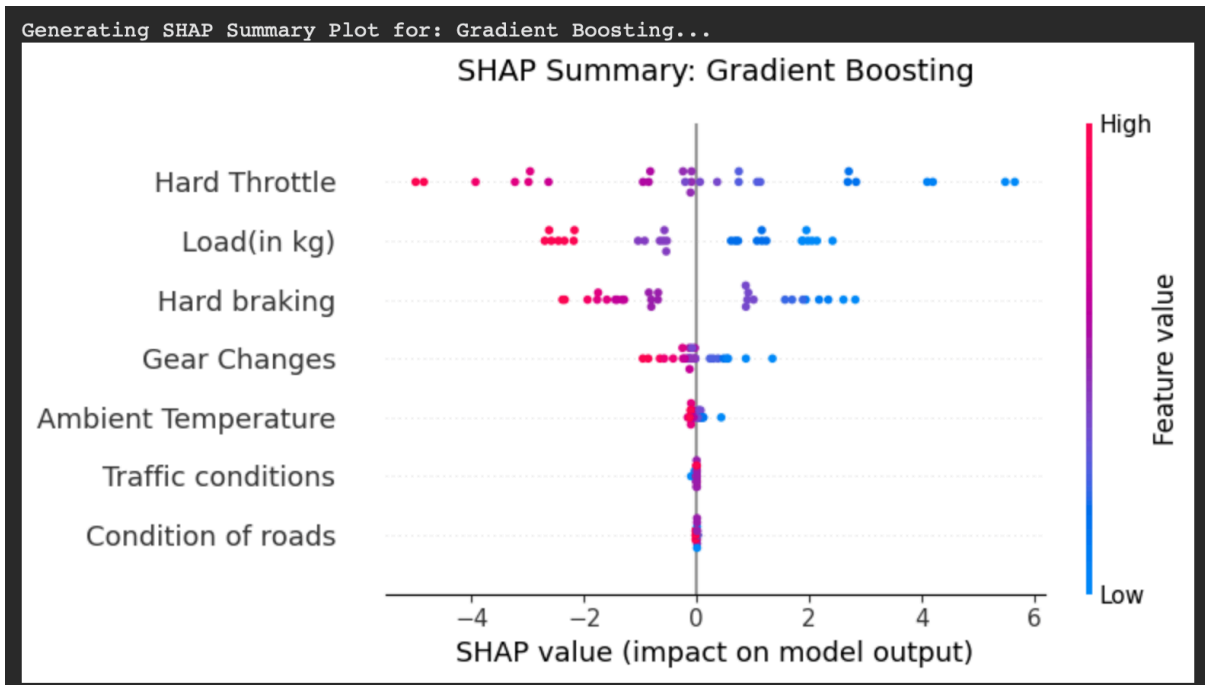


Fig.6 - The SHAP analysis result using the gradient boosting model(best model) is shown. The analysis shows that the hard throttle is the single greatest feature affecting the mileage. Also load shows a consistent negative correlation with the mileage.

11. Discussion & Conclusion

Our Synthesis Flow: This section synthesizes the quantitative metrics from our rigorous Leave-One-Out Cross-Validation (LOOCV) pipeline into qualitative physical realities. We evaluate whether our progressive modeling approach successfully answered the core problem statement, and how the inherent limitations of our 28-row real-world dataset affected the final physical interpretations.

11.1 Discussion

Interpretation of Results & Objectives Met: The primary objective of this study was to quantify the impact of human driving behavior versus environmental conditions on the fuel efficiency of a Harley-Davidson X440. We successfully met this objective. Our progression from Linear Models to Complex Ensembles and ultimately to pure Mathematical Curve Fitting successfully proved that fuel consumption is not a simple linear process; it behaves as a complex, non-linear function.

Insights into the Underlying Problem & Surprising Findings:

- **The "Squared" Reality of Mileage:** By comparing our ML models to a Polynomial Curve Fit (Degree 2), we proved that mileage drops at an exponential/squared rate under load and acceleration, rather than a steady linear decline.
- **The Dominance of Human Behavior:** A highly surprising finding confirmed by our SHAP analysis was the sheer mathematical dominance of Hard Throttle. Common assumptions often blame poor mileage on Traffic Conditions; however, our models revealed that the mechanical abuse of aggressive throttling completely overshadows the passive environmental drag of heavy traffic.
- **The Failure of Bagging on Small Data:** Another surprising technical finding was the failure of Random Forest during LOOCV testing (scoring the worst RMSE of 1.820). It demonstrated that "voting" architectures struggle to generalize when trained on synthetically interpolated data, whereas "correcting" architectures like Gradient Boosting succeed.

Impact of Small Dataset Size on Reliability: Our extreme data scarcity (N=28 real-world points) fundamentally shaped the reliability of our results. While our mathematical interpolation strategy successfully smoothed the models' decision boundaries, the reported Gradient Boosting R^2 of 0.994 is indicative of the "Hyper-Accuracy" statistical anomaly. Because interpolation has an "Information Cap," the model's reliability is strictly confined to the physical bounds of the testing day. While highly accurate within this window, the model cannot reliably extrapolate to edge cases (e.g., extreme weather or drastically different altitudes) not present in the original 28 points.

11.2 Conclusion

This project tackled the complex challenge of predicting motorcycle fuel efficiency using a highly constrained, primary ground-truth

dataset collected via physical "run-dry" testing of a Harley-Davidson X440. To overcome the severe statistical limitations of possessing only 28 distinct data points, the team engineered a robust pipeline featuring strict feature elimination via Variance Inflation Factor (VIF) analysis, pairwise data interpolation, and stringent Leave-One-Out Cross-Validation (LOOCV).

Among the evaluated approaches, Gradient Boosting emerged as the most capable machine learning predictor, achieving the lowest LOOCV RMSE (1.307) and successfully navigating the noise of the small dataset. However, the optimal mathematical explanation was provided by a Degree 2 Polynomial Curve Fit, which definitively proved that the underlying physics of fuel consumption act as a complex squared function rather than a simple linear equation.

Ultimately, the project goal was successfully achieved. By integrating advanced ensemble learning with SHAP interpretability, the study conclusively determined that aggressive human driving mechanics—specifically hard throttling—are the overwhelming primary factors degrading fuel efficiency, rendering external environmental conditions statistically marginal by comparison.

12. Learnings, Shortcomings & Future Work

Our Reflective Flow: A robust machine learning project must acknowledge not just what was achieved, but the friction experienced along the way. This section reflects on our technical growth, the honest limitations imposed by our physical testing, and our roadmap for scaling this architecture to the mass market.

12.1 Key Learnings

Our transition from classroom theory to applying ML on real-world, manually collected physical data yielded several critical learnings:

- **Learning 1 (Data Leakage & Multicollinearity):** We learned that raw physical data is inherently messy and interconnected. Identifying that Average Speed acted as an artificial cheat sheet (leakage) and utilizing VIF to ruthlessly prune redundant features like Gear Changes taught us that data quality strictly supersedes algorithm complexity.
- **Learning 2 (The Linearity Trap):** By testing a progressive suite of models, we learned firsthand the danger of the "Linearity Trap". We proved that assuming physical systems scale in straight lines is a critical engineering flaw, and that advanced architectures are required to map curved physical realities.
- **Learning 3 (Bagging vs. Boosting on Small Data):** We gained a profound practical understanding of tree ensembles. We learned that Bagging (Random Forest) struggles immensely to generalize when trained on synthetically augmented data, whereas sequential error-correction (Gradient Boosting) is vastly superior at isolating the true signal.

12.2 Shortcomings

While our mathematical validation pipeline was rigorous, we must acknowledge the physical limitations of our current implementation:

- **Shortcoming 1 (Extreme Data Scarcity):** The entire foundation of this model relies on 28 total data points. While our run-dry testing provided absolute ground truth, the sheer lack of volume remains the project's heaviest anchor.
- **Shortcoming 2 (The Information Cap):** To survive the scarcity, we relied on interpolation-based augmentation. However, synthetic generation has a hard "Information Cap"—it cannot invent physical variance or edge cases (e.g., a heavily

loaded bike in extreme uphill traffic) that were not present in the original 28 points, introducing a risk of synthetic bias.

- **Shortcoming 3 (The Hyper-Accuracy Anomaly):** Our highest performing model, Gradient Boosting, achieved an R^2 of 0.994. From an academic standpoint, we recognize this as a statistical anomaly indicative of overfitting to the specific test points, rather than a flawless mastery of motorcycle physics.

12.3 Future Work

Our current architecture proves that human driving behavior is the dominant variable in fuel efficiency. To evolve this from a restricted academic study into a scalable engineering solution, we propose the following future scope:

- **Longitudinal Data Strategy (12-Month Horizon):** Currently, our highly accurate model is based on a narrow time window. Future efforts require scaling data collection across a full year to capture variances across all four quarters (e.g., winter vs. summer ambient impacts, and long-term mechanical degradation).
- **High-Volume Commuter Motorcycle Target:** We intend to adapt this validated Gradient Boosting architecture for high-volume commuter motorcycles. By shifting our focus from premium bikes (Harley-Davidson X440) to mass-market models specifically like the **Hero Splendor**, we can optimize commuter efficiency at scale.
- **Automated Telemetry:** We plan to replace the manual "run-dry" physical fuel measurement with continuous, automated OBD-II (On-Board Diagnostics) scanner data to stream thousands of rows per trip directly into the data pipeline.

13. References

1. **Prof. Dilip Sharma (Dept. of Mechanical engineering, MNIT)** suggest to us the run dry method to collect the ground truth data.
2. **Harley Davidson X440:** <https://www.harley-davidsonx440.com>
3. **Library documentation:** scikit-learn. Machine Learning in Python (used for Linear, Ridge, Lasso, Random Forest, and Gradient Boosting implementations). <https://scikit-learn.org>
4. **Library documentation:** pandas & numpy. Python Data Analysis Libraries (used for dataset manipulation, VIF calculation, and synthetic interpolation). <https://pandas.pydata.org>
5. **Library documentation:** matplotlib & seaborn. Python Graphing Libraries (used for EDA visualizations, correlation heatmaps, and scatter plots). <https://matplotlib.org>
6. **Dataset:** Primary Manually Collected Dataset (`vehicle_mileage_dataset.csv`). Collected via physical "run-dry" testing on a Harley-Davidson X440. The dataset is attached in the Appendix-B.
7. **Online resource: Strava.** Route and Elevation Tracking Application. <https://www.strava.com> (utilized for gathering accurate trip telemetry, average speed, and elevation data during physical testing).
8. **Online resource:** Hero MotoCorp Official Website. Commuter Motorcycle Specifications. <https://www.heromotocorp.com> (utilized as the reference architecture for the mass-market scalability future scope targeting the Hero Splendor).

9. **Online resource:** Bajaj Motors Official Website. Motorcycle Specifications. <https://www.bajajauto.com>.
10. **Author(s).** (2017). Lundberg, S. M., & Lee, S. I. *A Unified Approach to Interpreting Model Predictions (SHAP)*. Advances in Neural Information Processing Systems 30. <https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions> (utilized for local and global model interpretability and feature importance analysis).

14. Code and Resources

14.1 Notebook

- Public Notebook Link (Google Colab): <https://colab.research.google.com/drive/1sBLkIXmX9RJHoJ3BWYQK33lZGlyjK8rX?usp=sharing>
- **Execution Status:** The notebook has been fully executed from top to bottom. All outputs, including the SHAP beeswarm plots, interpolation generation logs, and LOOCV evaluation metrics, are visible and preserved.
- **Access:** Link sharing has been set to "Public / Anyone with the link can view".

14.2 Code Repository

- GitHub Repository: <https://github.com/AkshatPurohit07/Mileage-prediction-using-ML>
- **Repository Contents:** The repository has been made public and contains:
 - The raw dataset (vehicle_mileage_dataset.csv).
 - The computer jupyter notebook link.
 - The jupyter notebook of the code in the pdf format.

- The presentation of this project.

14.3 Other Resources

- **Raw Dataset File:** The primary ground-truth dataset (`vehicle_mileage_dataset.csv`) is attached to this submission.
- **Telemetry Data Source:** Raw route, elevation, and average speed telemetry points were derived using the Strava mobile application during physical testing. Download link for the app from play store: <https://play.google.com/store/apps/details?id=com.strava> and on app store: <https://apps.apple.com/in/app/strava-run-bike-walk/id426826309>

Appendix

A. Complete Jupyter notebook in the pdf format:

https://drive.google.com/file/d/1U3jK8yd8aUMT9RceDU8pbhDrZdryvalA/view?usp=share_link

B. Final dataset used: <https://docs.google.com/spreadsheets/d/1P4MDfmxiPYVvkrsoFh1H5TyNjfFBg5cJ/edit?gid=1294632370#gid=1294632370>

C. The presentation of the project: <https://drive.google.com/file/d/1gkPlpw-EPXIFXQqbguL5aQEiZr9r63gO/view?usp=sharing>

D. Memories created during the project:

